

2025_Zebrafish_KO_data

June 8, 2025

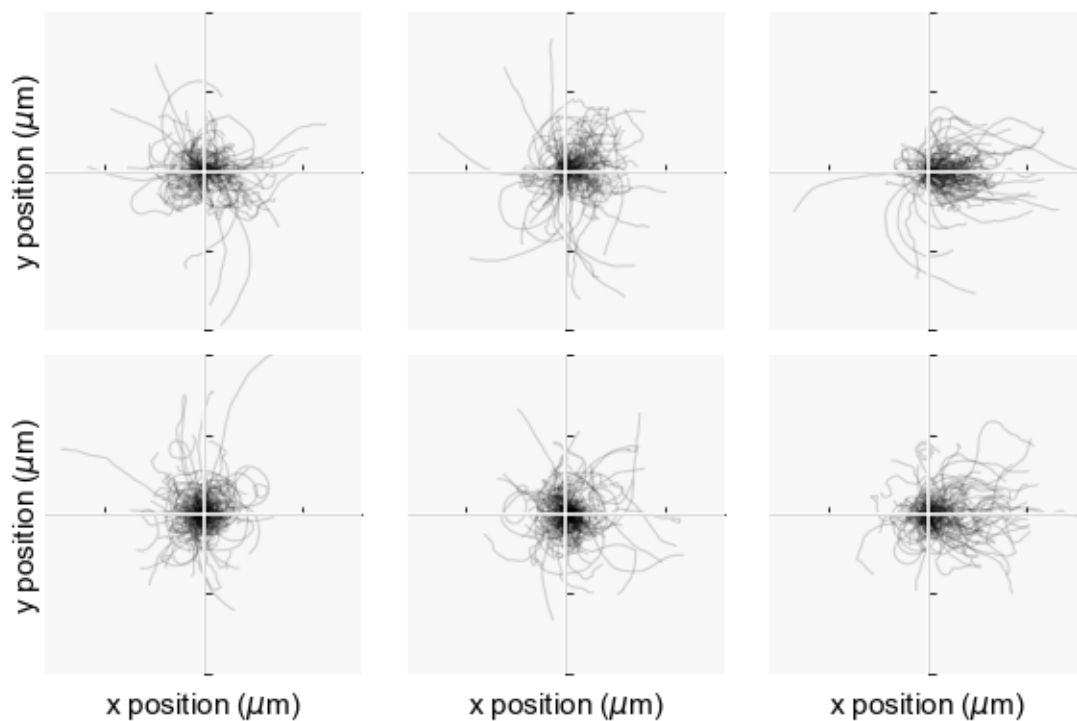
```
[2]: Unnamed: 0  cell  frame      x      y      z      celltype  \
0          0    0      0  349.024110  196.442253  0.0  Galvanin_WT
1          1    0      1  359.114003  194.286702  0.0  Galvanin_WT
2          2    0      2  369.991581  196.799344  0.0  Galvanin_WT
3          3    0      3  380.458097  202.806752  0.0  Galvanin_WT
4          4    0      4  383.005824  214.212646  0.0  Galvanin_WT

      position  E_V_cm      date  trial  user  celltype_genotype
0           2        0  20250403      0   AP      Galvanin_WT
1           2        0  20250403      0   AP      Galvanin_WT
2           2        0  20250403      0   AP      Galvanin_WT
3           2        0  20250403      0   AP      Galvanin_WT
4           2        0  20250403      0   AP      Galvanin_WT
```

1 Plotting

1.1 Tracks

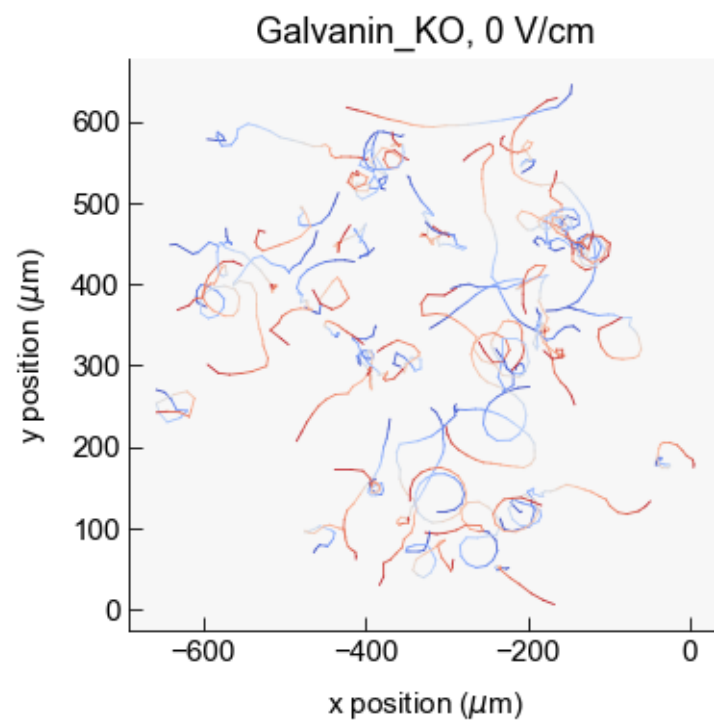
```
('Galvanin_KO', 0)
('Galvanin_KO', 1)
('Galvanin_KO', 3)
('Galvanin_WT', 0)
('Galvanin_WT', 1)
('Galvanin_WT', 3)
```



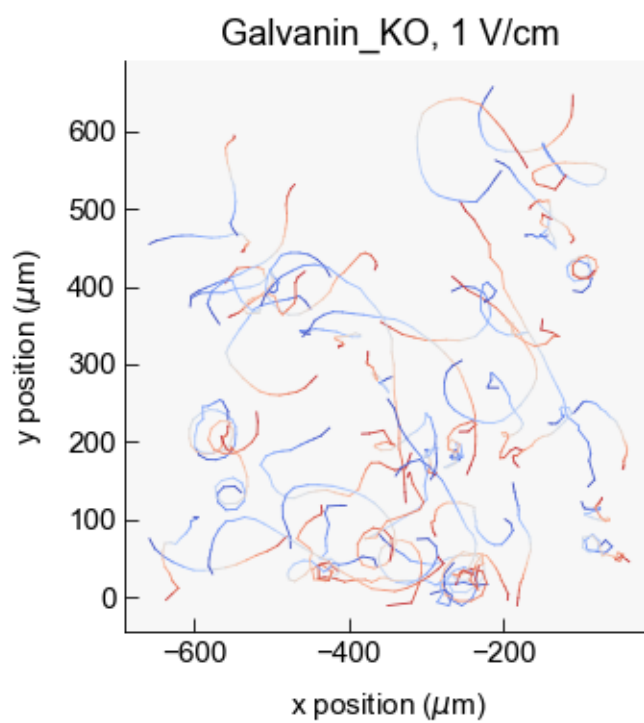
```

<>:53: SyntaxWarning: invalid escape sequence '\m'
<>:54: SyntaxWarning: invalid escape sequence '\m'
<>:53: SyntaxWarning: invalid escape sequence '\m'
<>:54: SyntaxWarning: invalid escape sequence '\m'
/var/folders/hs/8f67z23n6g9g9g4y03yhrn_c0000gn/T/ipykernel_60311/2643721333.py:5
3: SyntaxWarning: invalid escape sequence '\m'
    ax.set_xlabel('x position ($\mu$m)')
/var/folders/hs/8f67z23n6g9g9g4y03yhrn_c0000gn/T/ipykernel_60311/2643721333.py:5
4: SyntaxWarning: invalid escape sequence '\m'
    ax.set_ylabel('y position ($\mu$m)')
Processing group: ('Galvanin_KO', 0)

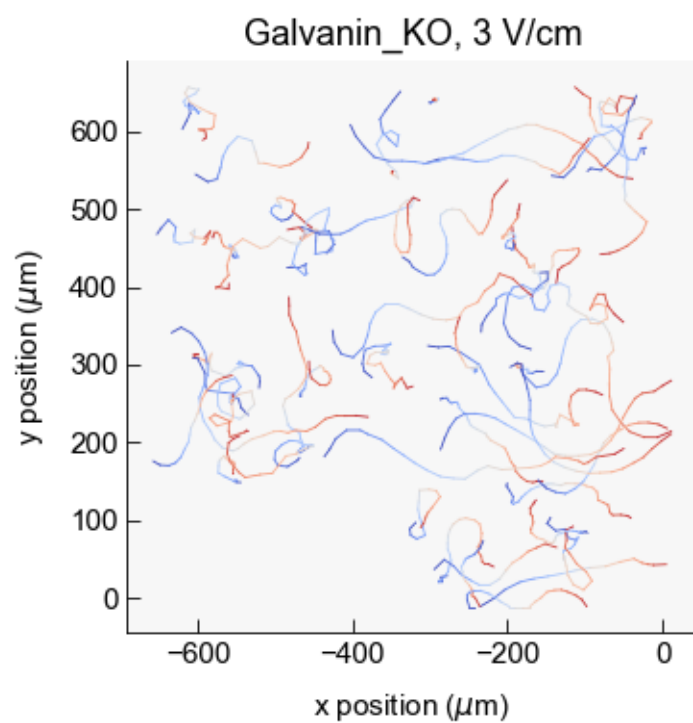
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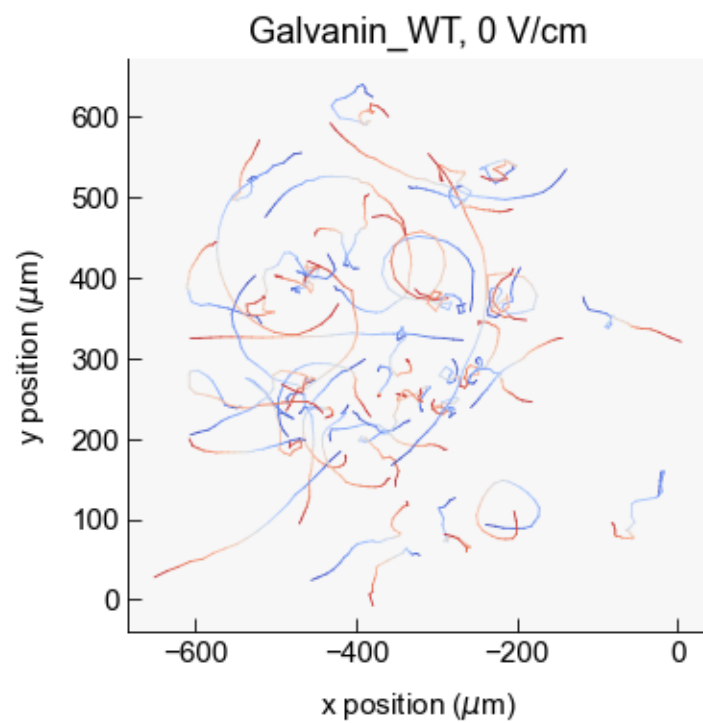
Processing group: ('Galvanin_KO', 1)



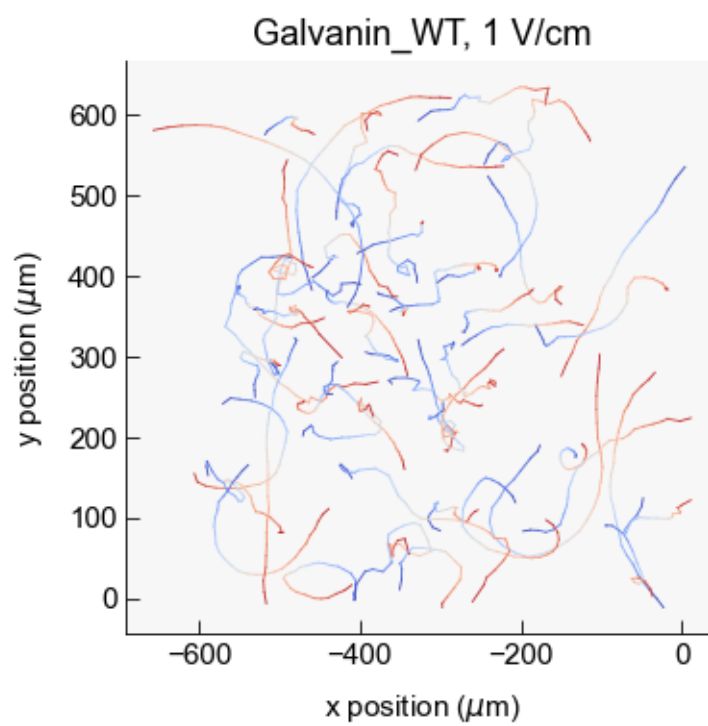
Processing group: ('Galvanin_KO', 3)



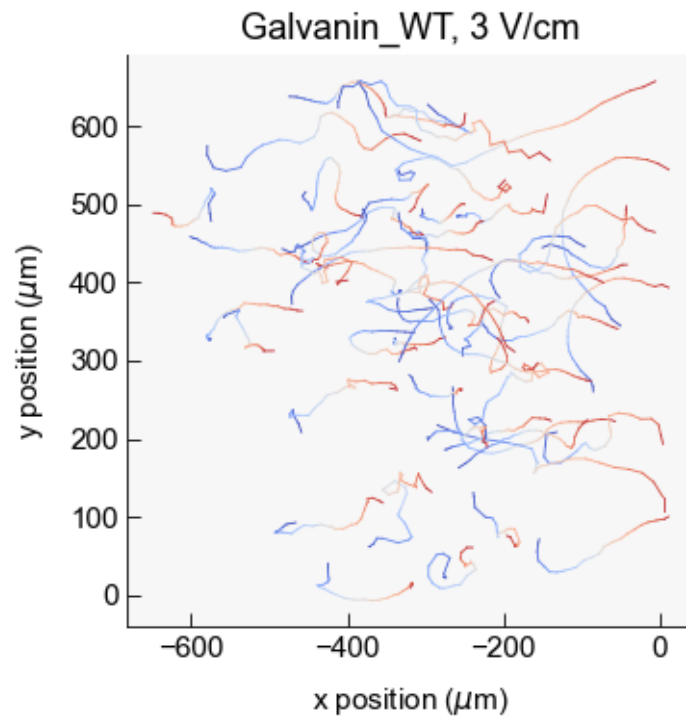
Processing group: ('Galvanin_WT', 0)



Processing group: ('Galvanin_WT', 1)



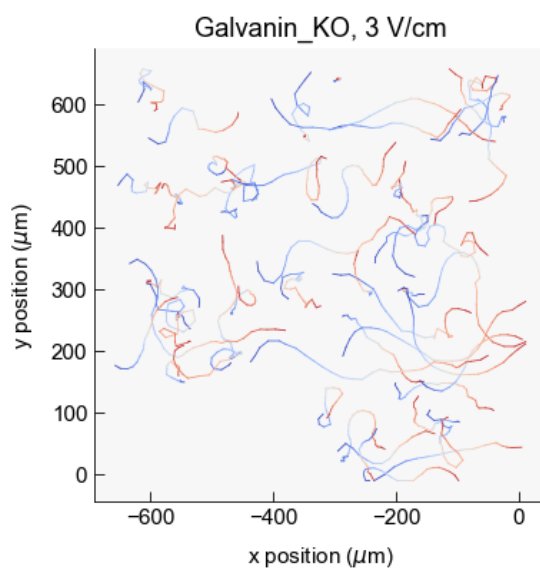
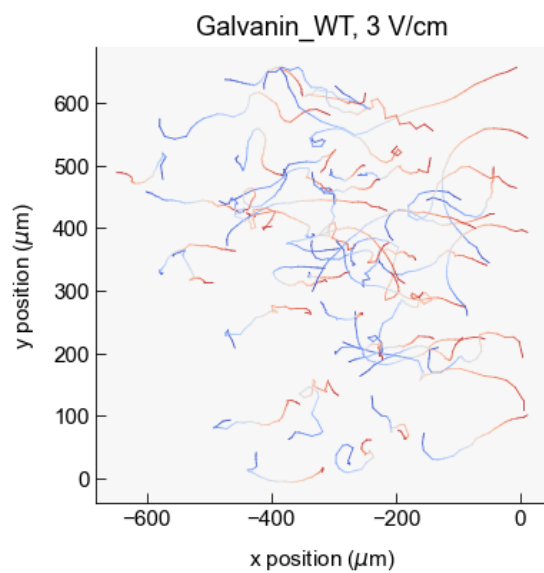
Processing group: ('Galvanin_WT', 3)



1.2 Here are only plots of 300 mV/mm condition only - probably more useful for main text figures. (two examples)

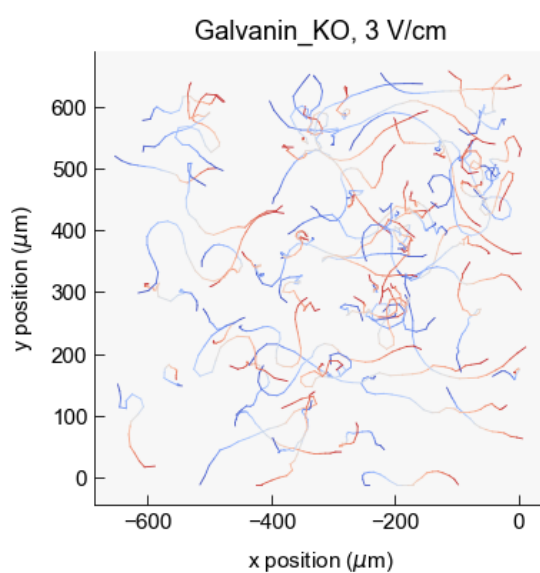
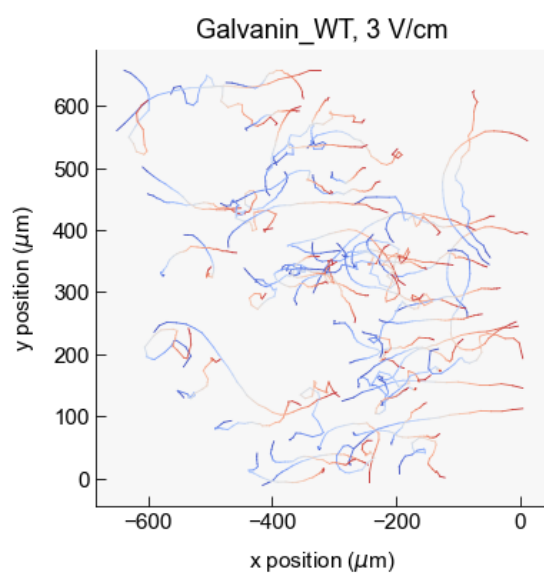
Processing group: ('Galvanin_KO', 3)

Processing group: ('Galvanin_WT', 3)

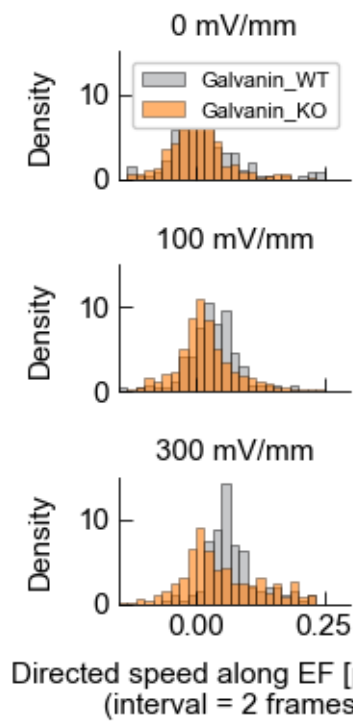
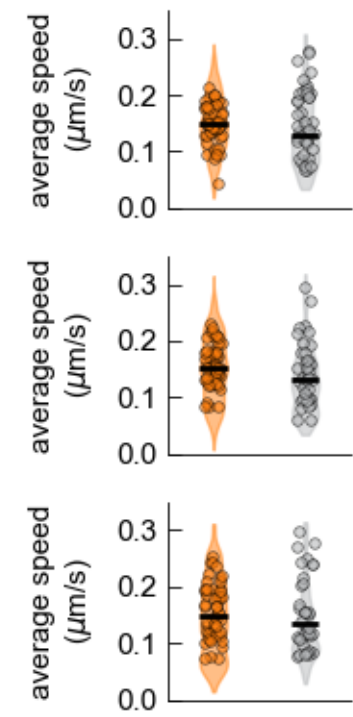


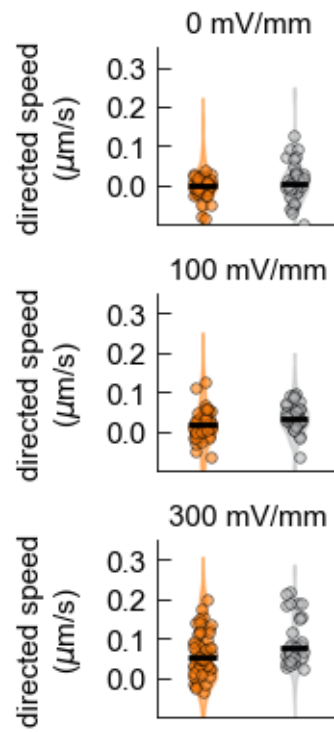
Processing group: ('Galvanin_KO', 3)

Processing group: ('Galvanin_WT', 3)

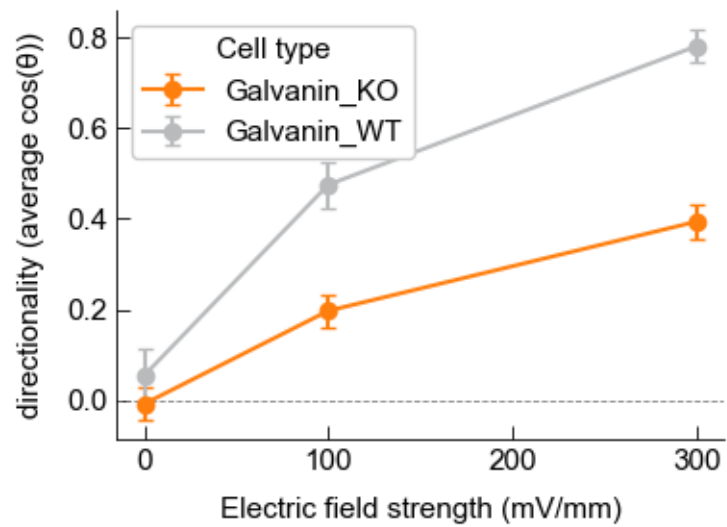


2 Speed related



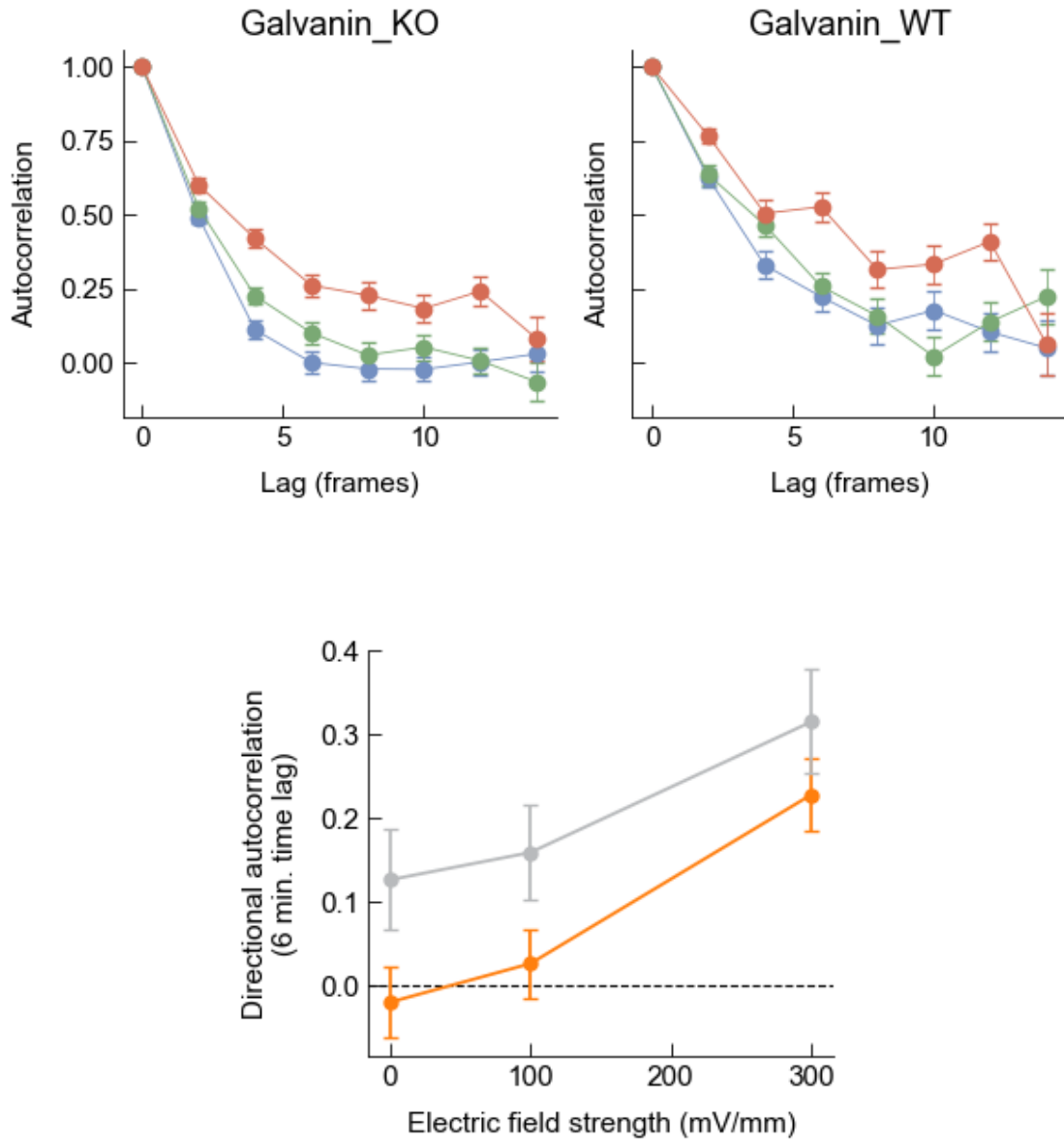


3 Directionality (average cosine theta)



4 Autocorrelation Analysis

4.1 Using cosine theta between trajectory vector and electric field.



4.2 Autocorrelation, but relative to 0 mV/mm condition.

Here we plot relative to the 0 mV/mm condition to try to account for residual persistence in the absence of an electric field. (i.e. subtraction of autocorrelation value from the 0 mV/mm condition).

